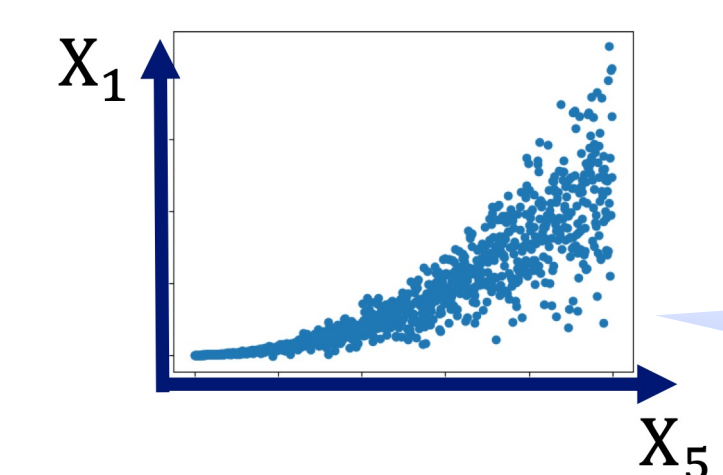
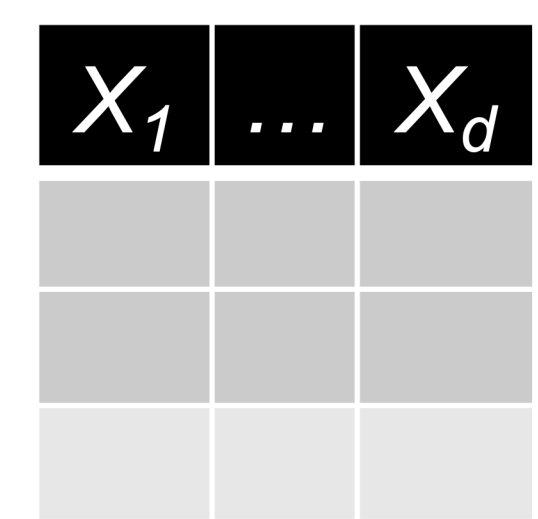


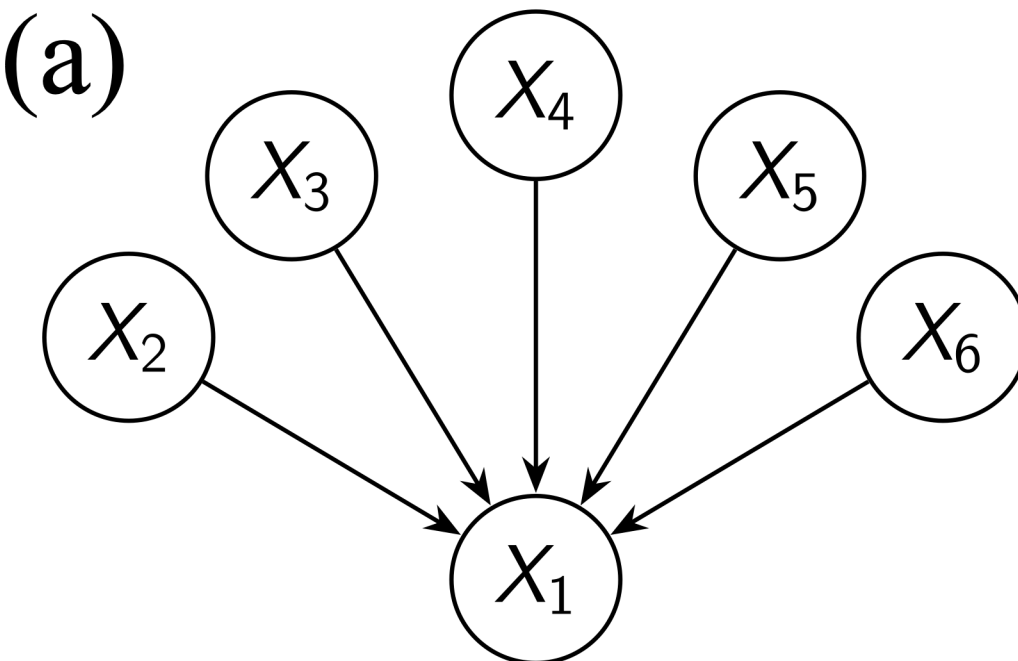
Standard Graphs vs. Mean & Variance Graphs: Which are more interpretable?

Observational data D

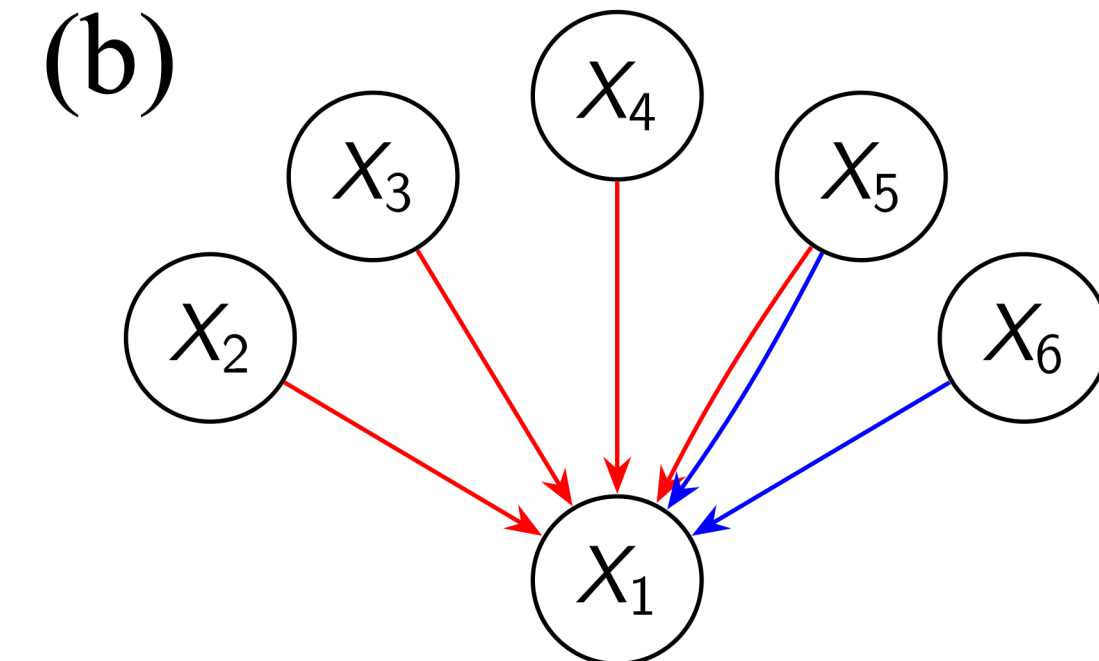


Heteroscedastic data

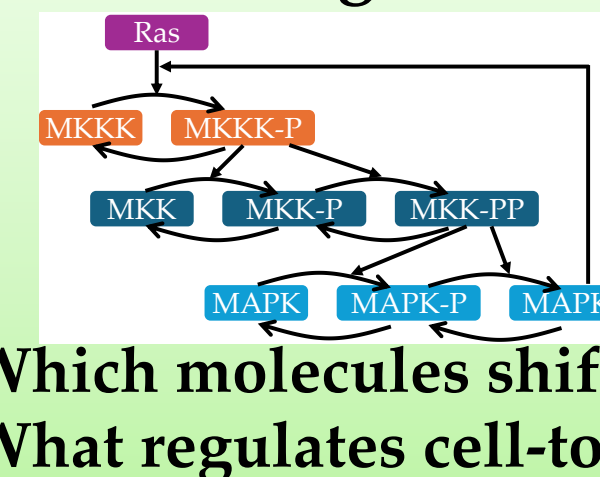
Standard causal graph



Mean/Variance causal graphs



Understanding in biology & medicine:



- Which molecules shift mean expression?
- What regulates cell-to-cell variability?

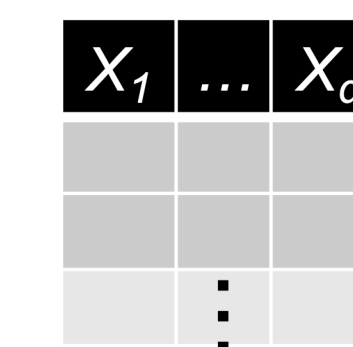
Latent discrimination detection in decision-making:



- Risk aversion by rejecting subgroups with high outcome variance is usual, but is it really fair?

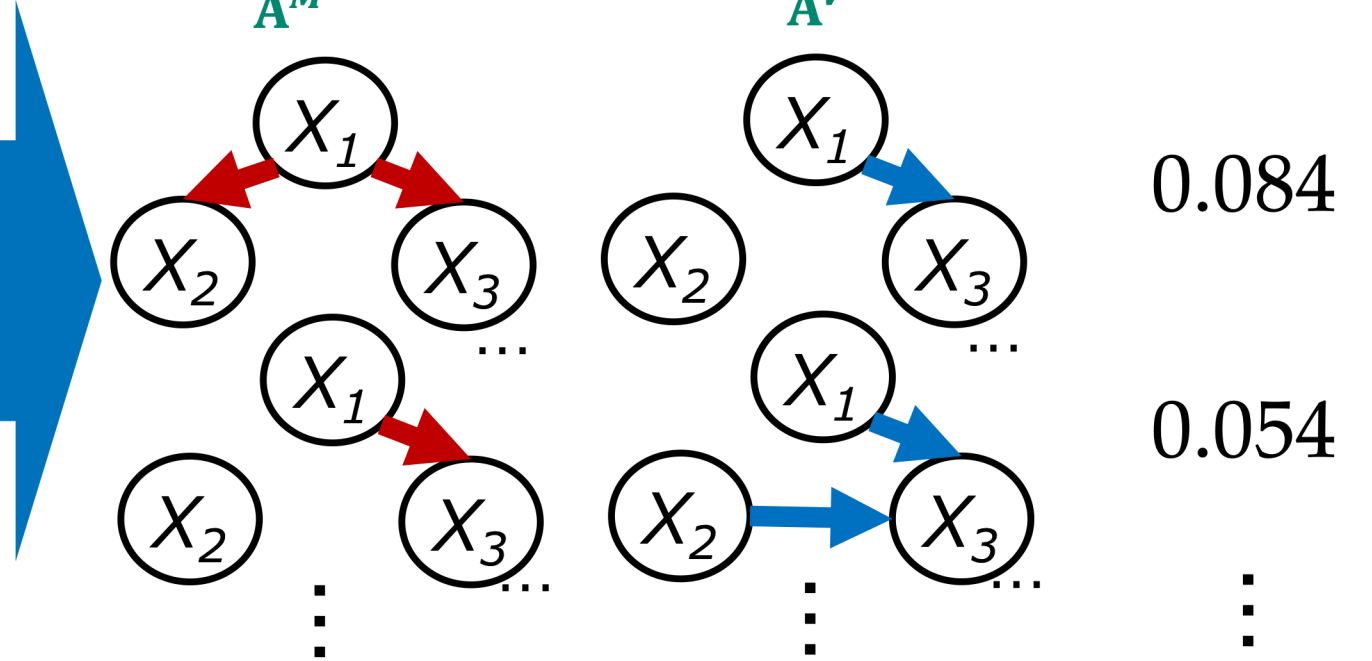
Setup: Bayesian Causal Discovery

Input Observational data D



Output Posterior over causal graphs $P(A^M, A^V | D)$

Mean causal graph Variance causal graph Probability



Input Prior knowledge about true causal graph structure

Edge of G^M Edge of G^V Order
 $X_1 \rightarrow X_3$ $X_1 \rightarrow X_3$ $\pi(X_1) < \pi(X_2)$

- Advantage: Can evaluate probability on structural feature of causal graphs

$$\mathbb{E}_{P(A^M, A^V | D)}[f(A^M, A^V)]; \text{ e.g., } f(A^M, A^V) = \mathbb{I}(X_i \rightarrow X_j \text{ in } A^M) \text{ or } \mathbb{I}(X_i \rightsquigarrow X_j \text{ in } A^V) \text{ (path)}$$

Models & Definition

Mean-variance HNMs: Each X_j 's values are determined by

$$X_j = m_j(X_{\text{pa}^M(j)}) + v_j(X_{\text{pa}^V(j)}) E_j \quad \text{for } j = 1, \dots, d,$$

mean function variance function zero-mean noise

$$\mathbb{E}[X_j | X_{\text{pa}(j)}] = m_j(X_{\text{pa}^M(j)})$$

$$\mathbb{V}[X_j | X_{\text{pa}(j)}] = \mathbb{V}[E_j] (v_j(X_{\text{pa}^V(j)}))^2.$$

Definition: Mean & Variance Causal Graphs

Mean causal graph G^M has edge $X_i \rightarrow X_j$ if and only if $X_i \in X_{\text{pa}^M(j)}$,

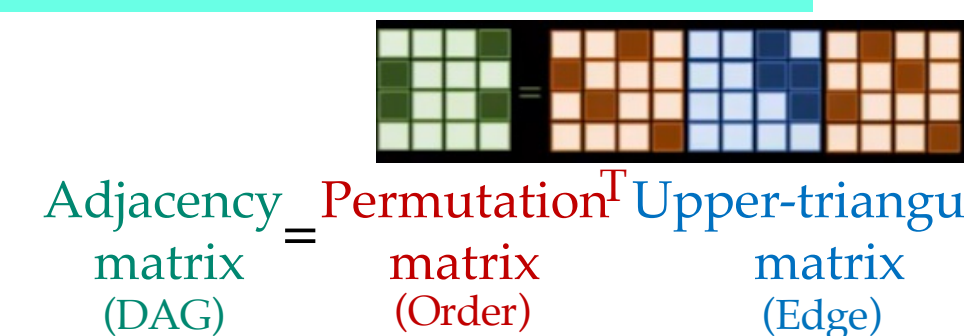
variance causal graph G^V contains edge $X_i \rightarrow X_j$ if and only if $X_i \in X_{\text{pa}^V(j)}$.

Adjacency Matrices: $A^M, A^V \in \{0, 1\}^{d \times d}$

Proposed Method

Goal Approximate the posterior as $P(A^M, A^V | \mathcal{D}) \approx P_\Phi(A^M, A^V)$

Identifiability Conditions Mean & Variance graphs are DAGs with shared permutation



$$A^M = \Pi^T U^M \Pi$$

$$A^V = \Pi^T U^V \Pi$$

$$\ast A_{i,j} = U_{\pi_i, \pi_j}$$

Factorized Variational Distribution $P(A^M, A^V) = \sum_{U^M, U^V, \Pi} P(U^M) P(U^V) P(\Pi)$

- Summation is computationally intractable $\otimes$
- Sampling operations for binary matrices U^M, U^V, Π are NOT differentiable $\otimes$

Differentiable sampling Approximate gradient using differentiable function

$$U^M \sim P_{\phi^M}(U^M)$$

$$U^V \sim P_{\phi^V}(U^V)$$

$$\Pi \sim P_\varphi(\Pi)$$

Gumbel-softmax sampling [Jang+; ICLR2016]:

Idea: Softmax(Unnormalized log prob. + Gumbel variables)

For each $i, j \in \{1, \dots, d\}$,

$$\tilde{U}_{i,j} \in [0, 1] = \frac{e^{(\log \phi_{i,j} + G_{i,j,a})/\tau}}{e^{(\log \phi_{i,j} + G_{i,j,a})/\tau} + e^{(-\log \phi_{i,j} + G_{i,j,b})/\tau}}$$

Gumbel-topK + SoftSort [Prillo+; ICML2020]:

Idea: SoftSort(Log prob. + Gumbel variables)

$$\hat{\pi} \in \mathbb{R}^d = \text{SoftSort}(\log \varphi + G) \rightarrow \hat{\Pi} \in [0, 1]^{d \times d}$$

Forward:
Discrete sampling
by taking argmax

Backward:
Use $\tilde{U}_{i,j} \in [0, 1]$ or $\hat{\Pi} \in [0, 1]^{d \times d}$
for gradient approximation

Objective Function $\max_{\Phi, \Theta} \mathbb{E}_{A^M, A^V \sim P_\Phi} [\log P_\Theta(\mathcal{D} | A^M, A^V)] - \lambda \Omega_{\Phi, \Theta},$

$$\log P_\Theta(\mathcal{D} | A^M, A^V) = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^d \left(\frac{(x_{i,j} - m_j(x_{i, \text{pa}^M(j)}; \theta_j^M))^2}{2(v_j(x_{i, \text{pa}^V(j)}; \theta_j^V))^2} + \log v_j(x_{i, \text{pa}^V(j)}; \theta_j^V) \right)$$

$$\Omega_{\Phi, \Theta} = \lambda_\Phi \text{KL}(P_\Phi(U^M, U^V) || P(U^M, U^V)) + \lambda_{\Theta^M} \sum_{j=1}^d \|\theta_j^M\|_2^2 + \lambda_{\Theta^V} \sum_{j=1}^d \|\theta_j^V\|_2^2,$$

Algorithm 1 Parameter Learning for Mean and Variance Causal Graphs

Require: Observational dataset $\mathcal{D} = \{x_1, \dots, x_n\}$; parameters Φ, Θ ; node-ordering constraints $\mathcal{O}$

1: Initialize all parameters $\Phi = \{\phi^M, \phi^V, \psi\}$ and $\Theta = \{(\theta_j^M, \theta_j^V)_{j=1}^d\}$

2: **while** Stopping criterion not met **do**

3: **while** Updates for parameters $\Theta \setminus \{(\theta_j^V)_{j=1}^d\}$ and $\Phi \setminus \{\phi^V\}$ not converged **do**

4: ① Sample mean and variance DAGs $A^M, A^V \sim P_\Phi(A^M, A^V)$

5: ② Compute ELBO-based objective function in Eq. (8)

6: ③ Update parameters $\Theta \setminus \{(\theta_j^V)_{j=1}^d\}$ and $\Phi \setminus \{\phi^V\}$ using scaled gradient

7: **if** node-ordering constraints $\mathcal{O}$ are provided **then**

8: Project updated parameters ψ onto the feasible set by Eq. (11)

9: **end if**

10: **end while**

11: **while** Updates for variance-related parameters $\{(\theta_j^V)_{j=1}^d\}$ and $\{\phi^V\}$ not converged **do**

12: ① Sample mean and variance DAGs $A^M, A^V \sim P_\Phi(A^M, A^V)$

13: ② Compute ELBO-based objective function in Eq. (8)

14: ③ Update variance-related parameters $\{(\theta_j^V)_{j=1}^d\}$ and $\{\phi^V\}$ using standard gradient

15: **end while**

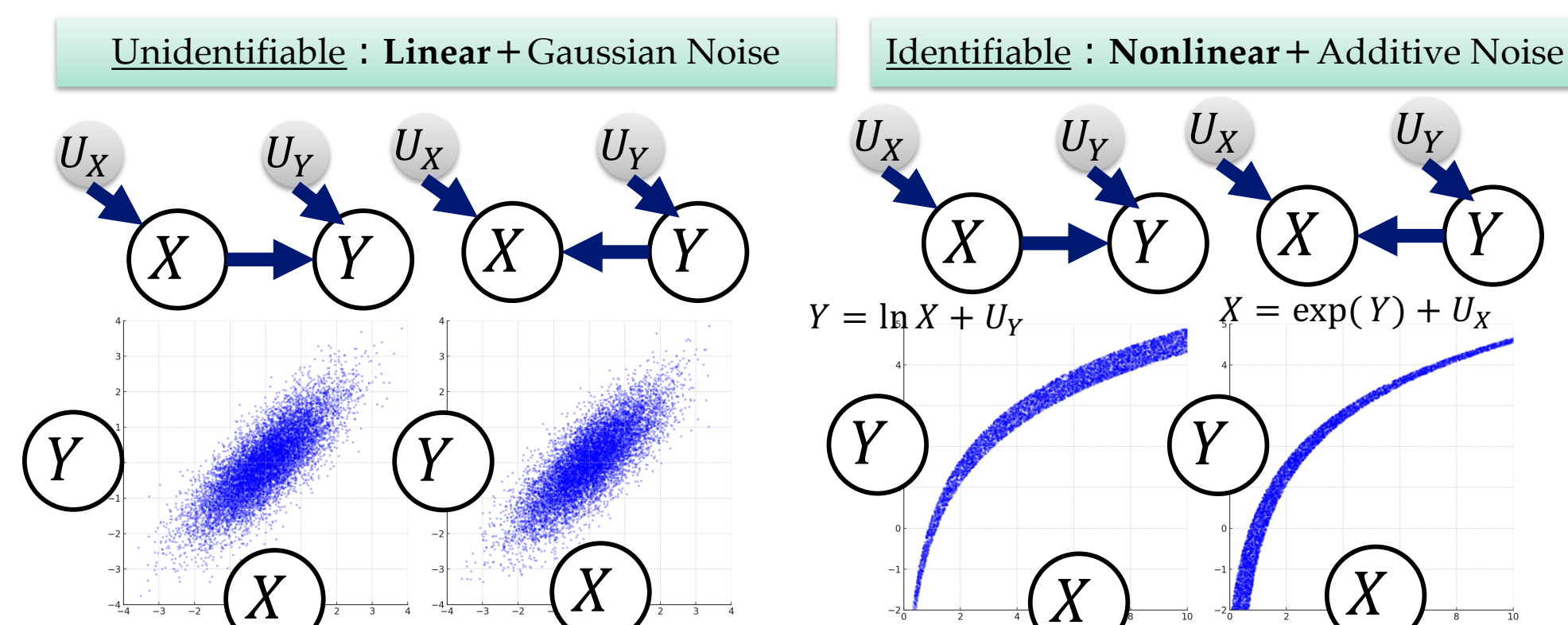
16: **end while**

17: **return** Learned parameters Φ and Θ

Order knowledge
 $\pi(X_1) < \pi(X_2)$

Identifiability Conditions

Unidentifiable = Different models can yield the same observational data distribution



Theorem 3.5: Under Assumptions 3.1, 3.2, 3.3, and 3.4, mean and variance causal graphs G^M and G^V are identifiable from observational distribution $P(\mathbf{X})$ if for $j = 1, \dots, d$, (A) m_j is a nonlinear function, (B) v_j is a piecewise function, **but not a constant function**, and (C) E_j is a Gaussian noise.

Assumption 3.1 (Causal sufficiency): Exogenous noises satisfy $E_i \perp E_j$ for any $i, j \in \{1, \dots, d\}$.

Assumption 3.2 (Causal minimality): $P(\mathbf{X})$ satisfies the causal minimality condition with respect to moment-agnostic causal graph G .

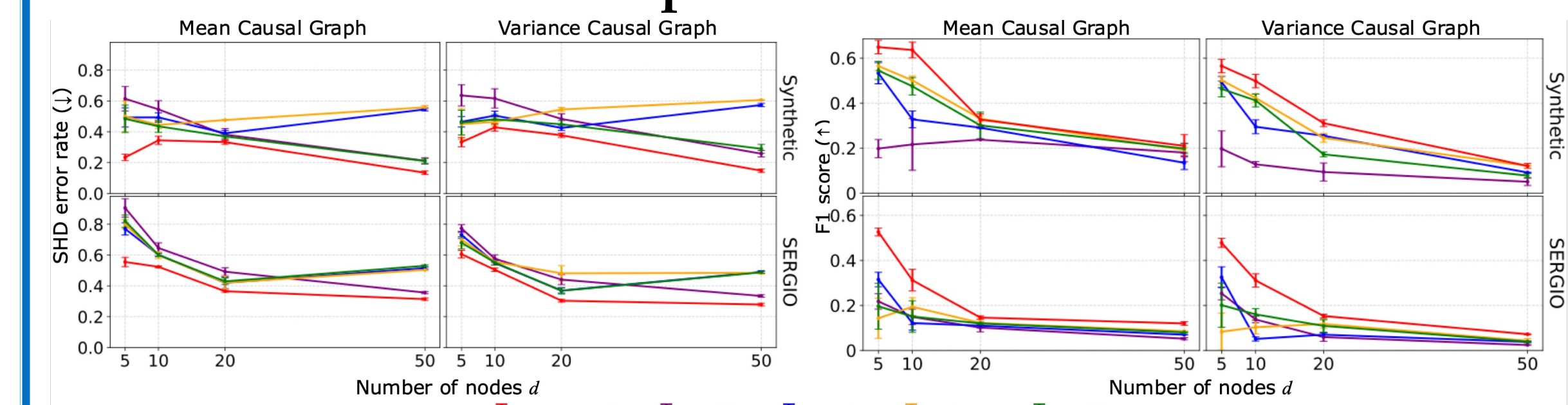
Assumption 3.3: Mean and variance graphs G^M and G^V are directed acyclic graphs (DAGs).

Assumption 3.4 (Shared permutation condition): There exists an identical permutation (a.k.a., topological ordering) of mean and variance causal graphs G^M and G^V .

Experimental Results

Synthetic Data & Semi-Synthetic Data (SERGIO Simulator):

✓ **Mean & Variance Graph Inference Results**



Effectiveness for moment-driven causal discovery:

Our method achieves lowest SHD & highest F1 score on both datasets.

Real-World Data (Sachs Dataset):

✓ **Standard Graph Inference Results**

	n = 100		n = 200		n = 853	
	SHD ($\downarrow$)	F1 ($\uparrow$)	SHD ($\downarrow$)	F1 ($\uparrow$)	SHD ($\downarrow$)	F1 ($\uparrow$)
MC3	22.6 $\pm$ 1.1	0.19 $\pm$ 0.04	20.3 $\pm$ 1.3	0.20 $\pm$ 0.07	18.9 $\pm$ 0.7	0.14 $\pm$ 0.05
DDS	17.6 $\pm$ 0.7	0.20 $\pm$ 0.03	16.6 $\pm$ 0.6	0.20 $\pm$ 0.02	15.0 $\pm$ 0.7	0.28 $\pm$ 0.02
ICDH	20.0 $\pm$ 1.0	0.21 $\pm$ 0.01	17.5 $\pm$ 0.5	0.22 $\pm$ 0.01	14.5 $\pm$ 0.5	0.27 $\pm$ 0.03
HOST	16.1 $\pm$ 0.5	0.19 $\pm$ 0.01	15.0 $\pm$ 0.3	0.33 $\pm$ 0.02	13.5 $\pm$ 0.8	0.38 $\pm$ 0.03
PROPOSED	16.0 $\pm$ 0.3	0.20 $\pm$ 0.02	14.9 $\pm$ 0.4	0.31 $\pm$ 0.02	13.7 $\pm$ 0.5	0.36 $\pm$ 0.02
PROPOSED +25%	14.9 $\pm$ 0.4	0.34 $\pm$ 0.03	14.8 $\pm$ 0.5	0.35 $\pm$ 0.02	13.4 $\pm$ 0.4	0.37 $\pm$ 0.02
PROPOSED +50%	13.2 $\pm$ 0.5	0.36 $\pm$ 0.02	13.2 $\pm$ 0.3	0.36 $\pm$ 0.02	13.1 $\pm$ 0.2	0.45 $\pm$ 0.03

Comparable moment-agnostic causal discovery performance:

Order knowledge incorporation effectively addresses data scarcity.

✓ **Case Study: Variance-Level Edge Detection**

Table 4: Estimated posterior probability of MEK $\rightarrow$ ERK in inferred variance causal graph structure

	n = 100	n = 200	n = 853
PROPOSED	0.585 $\pm$ 0.009	0.592 $\pm$ 0.008	0.620 $\pm$ 0.019

High posterior probabilities in low-sample regimes:

Consistent with biological evidence, **our method** successfully discovers the drivers of heteroscedasticity.

References:

1. Eric Jang, Shixiang Gu, and Ben Poole. *Categorical Reparametrization with Gumbel-Softmax*. ICLR, 2017.
2. Sebastian Prillo and Julian Eisenschlos. *Softsort: A Continuous Relaxation for the Argsort Operator*. ICML, 2020.

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